

SINAN, China

WMO: 577310

Lat: 27.9311N

Lon: 108.2450E

Elev: 1821

StdP: 13.75

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	33.0	35.4	21.8	17.5	40.8	24.9	20.3	39.4	8.3	47.3	7.5	46.8	2.3	0	0.311

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.0	96.1	76.1	94.0	75.9	92.0	75.5	79.0	89.8	78.2	88.5	77.4	87.1	5.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.4	147.7	83.0	75.7	143.7	82.3	74.9	140.3	81.8	44.3	90.1	43.4	88.8	42.5	87.2	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.6	7.8	6.8	32.1	99.5	2.3	1.8	30.5	100.8	29.1	101.8	27.9	102.8	26.2	104.1		
(5)			29.7	80.4	2.4	1.2	28.0	81.2	26.6	81.9	25.2	82.6	23.5	83.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	64.9	44.4	48.4	56.0	65.4	72.2	77.3	82.9	82.6	76.3	66.4	57.6	47.9
	DBStd	14.40	5.78	7.01	7.30	6.95	5.16	4.90	4.19	4.51	6.01	5.76	5.83	4.76
	HDD50	423	189	104	24	1	0	0	0	0	0	0	9	96
	HDD65	2297	638	466	293	79	4	0	0	0	2	52	232	531
	CDD50	5848	16	60	211	462	687	820	1019	1011	789	507	236	30
	CDD65	2245	0	2	14	89	226	370	554	546	341	94	9	0
	CDH74	20539	1	17	100	575	1431	2967	6240	5935	2820	424	29	0
	CDH80	8636	0	3	23	169	482	1091	2884	2756	1123	101	4	0
Wind	WSAvg	2.6	2.2	2.4	2.6	2.8	2.7	2.8	3.4	3.2	2.7	2.2	2.2	2.1
Precipitation	PrecAvg	44.8	1.1	1.1	2.2	4.3	6.5	7.9	7.0	4.7	3.1	3.9	2.0	0.9
	PrecMax	55.8	2.8	2.7	5.2	8.4	9.0	13.7	17.2	10.3	6.0	7.6	4.6	2.8
	PrecMin	32.6	0.2	0.1	0.7	1.1	3.2	3.0	1.0	1.3	0.2	1.8	0.2	0.0
	PrecStd	5.8	0.6	0.7	0.9	1.9	1.6	2.8	4.1	2.4	1.3	1.6	1.1	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.3	76.1	83.2	89.2	92.6	94.4	98.3	99.2	96.3	88.0	77.5	63.9
		MCWB	53.6	60.7	66.1	70.3	72.2	76.2	76.6	75.7	74.6	71.1	65.3	54.3
	2%	DB	58.4	66.7	76.3	84.9	89.0	91.8	96.0	96.5	92.6	82.6	72.6	60.0
		MCWB	50.3	56.2	63.1	68.7	70.7	75.5	76.7	75.9	74.2	68.0	61.9	52.4
	5%	DB	54.7	61.6	71.2	80.8	85.5	89.3	93.9	94.2	90.0	78.7	68.8	56.8
		MCWB	48.1	53.3	59.8	67.3	70.4	75.2	76.7	76.0	73.6	67.2	60.4	50.2
	10%	DB	52.1	57.6	66.7	76.4	81.9	86.5	91.7	91.7	86.7	75.1	65.5	54.5
		MCWB	46.8	51.3	57.9	65.6	69.8	74.8	76.5	75.8	72.7	66.0	59.1	48.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.9	61.4	67.9	73.5	76.6	79.1	80.4	80.0	77.6	73.2	67.1	57.2
		MCDB	62.8	74.7	80.9	84.3	86.3	89.6	91.0	91.2	89.6	82.9	76.2	61.9
	2%	WB	51.3	57.6	64.2	71.1	74.6	77.9	79.2	78.9	76.4	70.9	63.9	53.6
		MCDB	56.0	65.4	74.0	80.7	83.3	87.5	89.5	90.3	87.3	78.2	69.2	57.8
	5%	WB	49.3	54.5	61.4	68.9	73.0	76.8	78.3	78.0	75.4	69.0	62.0	51.6
		MCDB	53.5	60.0	69.0	77.6	81.1	85.1	88.4	88.6	85.5	75.6	67.1	55.9
	10%	WB	47.4	52.2	59.0	66.9	71.5	75.9	77.5	77.2	74.3	67.0	59.9	49.6
		MCDB	51.7	57.0	65.5	74.6	78.6	83.6	87.3	86.7	83.7	73.0	64.6	53.7
Mean Daily Temperature Range	5% DB	MDBR	7.5	9.3	11.7	13.3	13.3	12.0	14.0	15.0	13.4	10.2	9.6	7.9
		MCDBR	14.6	18.8	21.4	22.7	21.9	18.1	18.4	19.3	19.2	17.8	16.8	13.9
		MCWBR	7.8	9.3	9.2	8.4	6.6	5.1	3.8	3.9	4.5	5.8	7.2	7.1
	5% WB	MCDBR	13.3	17.1	18.5	18.3	15.2	13.7	15.6	16.4	16.2	13.6	13.7	11.6
		MCWBR	7.5	9.1	8.8	8.2	6.3	4.6	4.3	4.4	4.5	5.7	6.6	6.6
Clear-Sky Solar Irradiance	taub	0.654	0.668	0.730	0.698	0.603	0.579	0.527	0.574	0.602	0.682	0.673	0.665	
	taud	1.520	1.525	1.462	1.568	1.813	1.910	2.069	1.914	1.803	1.571	1.544	1.509	
	Ebn at Noon	181	196	195	209	231	235	247	235	222	190	176	170	
	Edh at Noon	79	84	95	88	69	62	53	61	67	79	76	77	
All-Sky Solar Radiation	RadAvg	454	591	828	1018	1099	1118	1494	1392	1096	733	620	468	
	RadStd	104	142	105	161	195	182	143	142	143	122	125	104	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.54	N/A	N/A	+0.92	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	N/A	+1.08	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page